Atul Kaple

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Professional Summary

Forward Deployed Engineer specializing in building high-concurrency systems and AI-assisted workflows with TypeScript, React, and Node.js. Delivers mission-critical automation and cloud integrations for enterprise clients.

Technical Skills

Core Languages: TypeScript, JavaScript, Python, Java
Frontend Technologies: React.js, Next.js, Tailwind CSS, Redux, Zustand
Backend & APIs: Node.js, Express.js, REST APIs, Socket.io, PostgreSQL, MongoDB
Cloud & DevOps: AWS (S3, CloudFront, Elastic Beanstalk), Docker, Git & GitHub, Redis

Experience

SidLabs LLP **Nov 2024 – Present**
Full Stack Developer *Delhi, India (Remote)*

Legal Case Automation (Next.js, Node.js, Google Gemini)

- Built a case evaluation flow on Google Gemini that checks Swiss asylum admissibility and drafts structured UN Human Rights appeals (CAT, CEDAW) of 3,000+ words, cutting case turnaround by 50%.
- Wrote browser-side text extraction with Tesseract.js, pdfjs-dist and Mammoth that reads PDFs, scanned images and Word files without a server round trip, replacing manual evidence transcription.
- Generated court-ready .docx petitions from case data with Docxtemplater and PizZip, sorting evidence into standard annexes automatically.

Patient Community Network (React, TypeScript, Node.js, Socket.io, Redis)

- Developed a privacy-focused social network with polls, media uploads, mood tags, anonymous posting and a ranked feed; Redis caching and cursor pagination support 2,000 concurrent users.
- Indexed posts and comments for semantic search with Transformers.js and MongoDB Atlas Vector Search (RAG), returning summaries that link back to the original posts.
- Gave authors audience controls and applied HIPAA Safe Harbor rules to strip identifying details before posts reach clinical researchers.

Healthcare Research Application (React, TypeScript, Node.js, AWS)

- Created a five-role system (patient, caregiver, clinician, researcher, admin) with an HL7 FHIR R4 assessment module, MongoDB transactions, automatic scoring and S3 video streaming.
- Met HIPAA and GDPR requirements with tiered consent checks, separate MongoDB clusters for health and public data, AWS Cognito MFA and a SHA-256 hash-chained audit log.
- Wrote a de-identification job (AWS KMS decryption, HMAC-SHA256 pseudonyms, small-group suppression with $k < 5$) over 34+ datasets; synced Apple HealthKit data and scheduled medication reminders.

Multi-Tenant Enterprise Platform (Next.js 15, TanStack Query, Socket.io, Razorpay, ExcelJS)

- Assembled a multi-tenant management portal with Next.js 15, React 19, TypeScript and Node.js/Express, isolating dual-database connections to serve corporate clients with zero cross-tenant contamination.
- Slashed network payload sizes by 72% and achieved sub-10ms cached view paints by deploying offline-first TanStack React Query persistence with client-scoped localStorage caching and HTTP gzip compression.
- Cut redundant database roundtrips by 90% through background sync-status hashing, and dropped dashboard latency from 1,400ms to $<$ 45ms (96% reduction) using in-memory TTL caching and compound indexes.

Environmental Monitoring Dashboard (Next.js, Leaflet, Node.js, MongoDB)

- Compressed multi-GB environmental datasets into a 4.4MB JSON snapshot through schema encoding and truncation, an 80% size cut that brought dashboard load time under one second.
- Scheduled a node-cron job with Compromise NLP that reads 30+ RSS feeds every 15 minutes, matches 120+ threat keywords, geocodes 200+ Peruvian regions via Nominatim and removes duplicates in MongoDB.
- Drew Leaflet maps with Turf.js 20km buffers and point-in-polygon tests on protected reserves to flag nearby threats and rank the top 5 deforestation hotspots.

Personal Project

Collaborative Whiteboard Application (Next.js, TypeScript, Liveblocks) | Live Link | Demo

- Architected an infinite-canvas collaborative workspace supporting 16+ vector primitives, sticky notes and connector arrows using Next.js 14, React 18, TypeScript and Tailwind CSS.
- Implemented real-time CRDT synchronization via Liveblocks (LiveMap/LiveList) with 16ms throttling (60 FPS), powering conflict-free multi-user editing, ephemeral cursor chat, live reactions, and camera spotlight follow mode.
- Engineered client-side optimizations and secure architecture, utilizing the Ramer-Douglas-Peucker (RDP) algorithm for stroke point reduction, and Clerk + Convex for multi-tenant RBAC with sub-100ms reactive database queries.

Education

Army Institute of Technology **June 2020 – May 2024**
Bachelor of Engineering, Mechanical Engineering | GPA: 8.0 *Pune, Maharashtra*